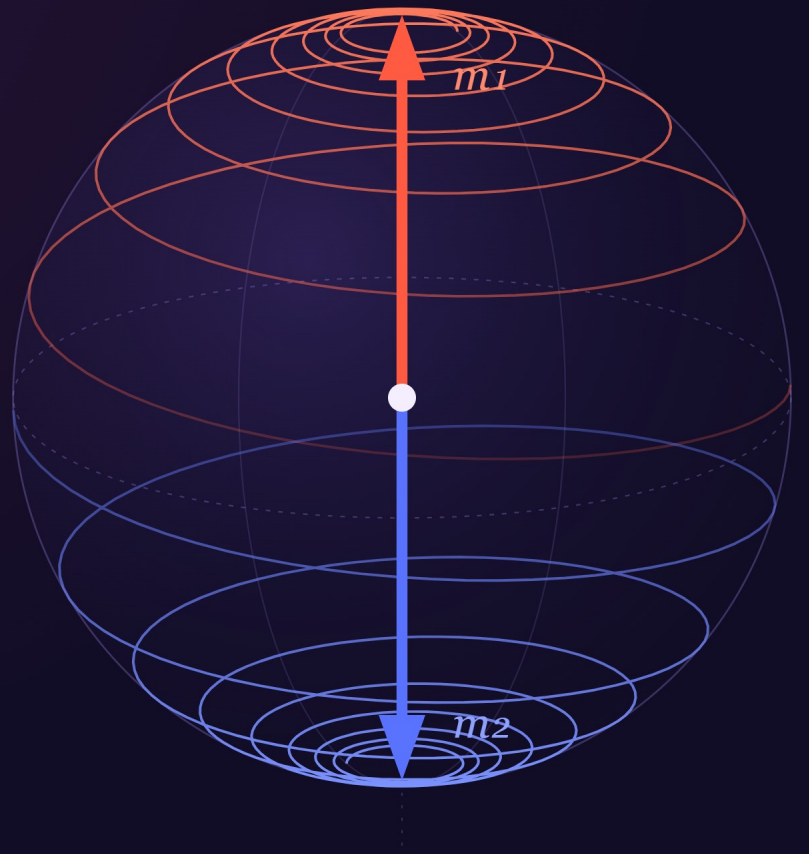
spinmate

AI AGENT FOR SPINTRONICS SIMULATION

# Spintronics simulation, by conversation.

Describe a setup in plain language or provide a parameter file. SpinMate checks it with the solver's own parser, runs two-sublattice macrospin simulations and explains the result from the actual magnetization dynamics.

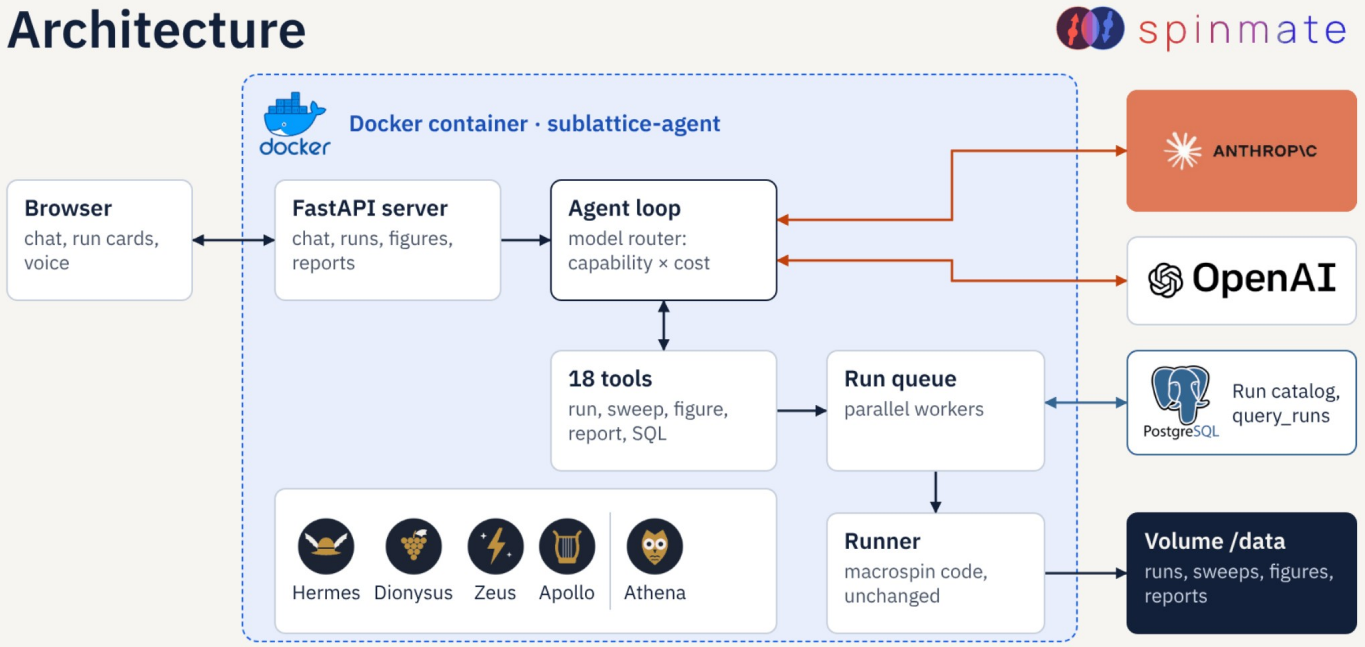



Michael Lianeris · github.com/Michaellianeris/sublattice-agent · MIT license

# Outline


| | | |
|---|---|---|
| 01 | **Architecture** | How the browser, the agent, the tools and the solver fit together |
| 02 | **Models** | Five personas over Claude and OpenAI, and how the router chooses |
| 03 | **Value** | Runs, sweeps, figures, reports and the catalog, on real runs |
| 04 | **In numbers** | Speed, tests, tools and the codebase |
| 05 | **How to run** | Docker, the Windows app and the first start |

# Architecture




### In the browser

Chat, run cards, Runs panel, voice input

### In the container

FastAPI server, agent loop with the router, 18 tools, run queue, runner

### Outside it

Claude and OpenAI, the run catalog, the /data volume

# Five personas, one router



## Hermes

**Quick and economical**

Claude Haiku, GPT nano / mini



## Dionysus

**Balanced, the default**

Claude Sonnet, GPT-5.x



## Zeus

**Deep, most capable**

Claude Opus, OpenAI o-series



## Apollo

**Specialised and new**

Claude Fable / Mythos, new models



## Athena

**The guide**

How to use the app; built-in answers offline

**ON AUTO, EVERY REQUEST IS SCORED OVER THE PERSONA'S MODELS**

$$\text{score} = \text{capability} + \text{bonus} - \lambda(\text{persona}, \text{request}) \times \log_{10}(\text{price} / \text{cheapest}) - \text{shortfall}$$

The strongest model wins unless a cheaper one is clearly enough. Every answer shows the model, the reason and the cost in €, at the daily ECB rate.

# What SpinMate gives you

**BY HAND**

1. Edit the parameter file
2. Run the solver and wait
3. Write a plotting script
4. Read the curves for switches and settling
5. Collect parameters and figures into a write-up

**WITH SPINMATE**

1. One sentence, or the same .txt file
2. Every value checked by the solver’s parser
3. Run card with the sphere, curves and metrics
4. Answer written from the computed metrics
5. IEEE figure and PDF report on request

**Same solver, same physics: the agent removes the steps around it, not the model.**

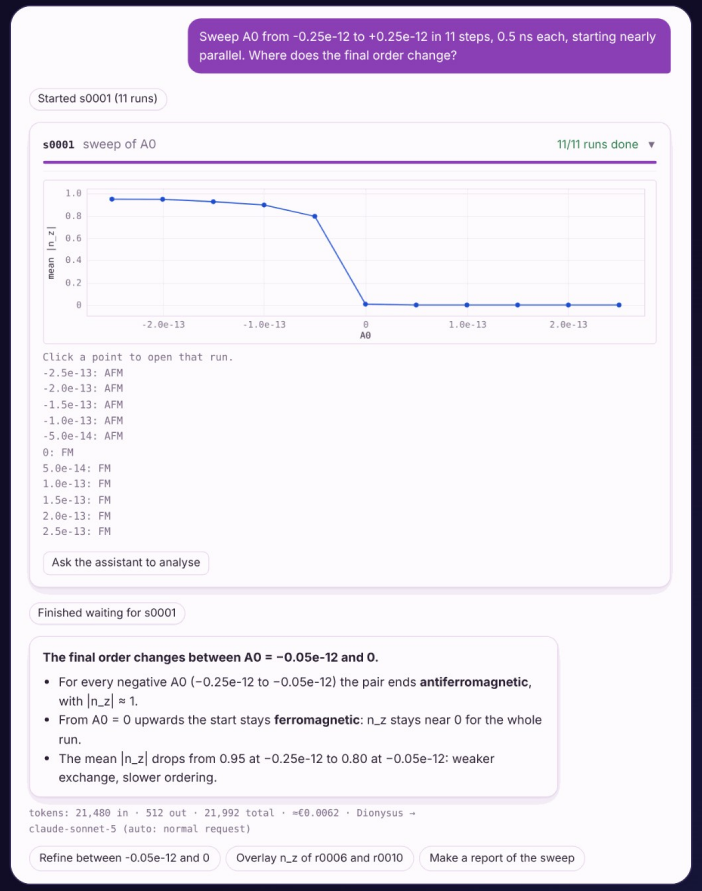
# One question, one run




## Behind one question

1. The router picks a model for the chosen persona and request.
2. The model calls validate_parameters, then start_simulation.
3. The run queue starts the solver; the card shows live progress.
4. wait_for_run returns the metrics, and the answer is written from them.

# Sweeps, thresholds and continued runs



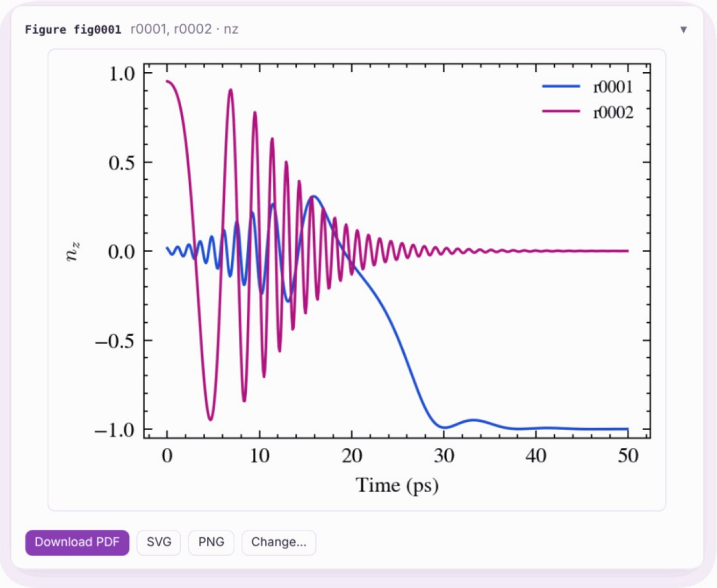

*“Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each. Where does the final order change?”*

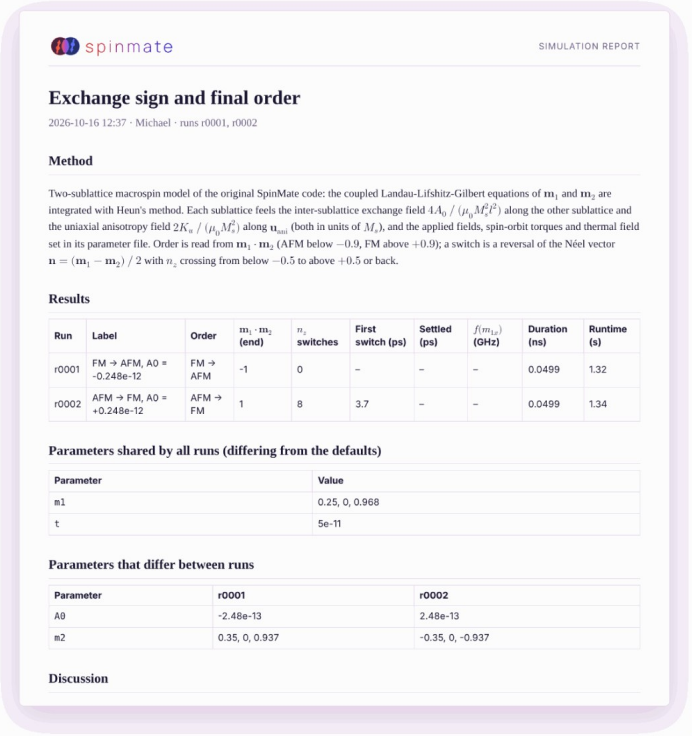
RESULT

## The order changes between A0 = −0.05e-12 and 0: AFM below, FM from 0 upwards.

- Runs go in parallel; click a point to open that run.
- Threshold search: bracketing sweeps narrow the interval.
- Continue run keeps the clock, so AC fields and the FeRh cycle do not restart.

# Figures and reports, ready to share





spinmate

SIMULATION REPORT

**Exchange sign and final order**

2026-10-16 12:37 · Michael · runs r0001, r0002

**Method**

Two-sublattice macrospin model of the original SpinMate code: the coupled Landau-Lifshitz-Gilbert equations of $\mathbf{m}_1$ and $\mathbf{m}_2$ are integrated with Heun's method. Each sublattice feels the inter-sublattice exchange field $4A_0 / (\mu_0 M_s^2 l^2)$ along the other sublattice and the uniaxial anisotropy field $2K_u / (\mu_0 M_s^2)$ along $\mathbf{u}_{ani}$ (both in units of $M_s$), and the applied fields, spin-orbit torques and thermal field set in its parameter file. Order is read from $\mathbf{m}_1 \cdot \mathbf{m}_2$ (AFM below $-0.9$, FM above $+0.9$); a switch is a reversal of the Néel vector $\mathbf{n} = (\mathbf{m}_1 - \mathbf{m}_2) / 2$ with $n_z$ crossing from below $-0.5$ to above $+0.5$ or back.

**Results**

| Run | Label | Order | $\mathbf{m}_1 \cdot \mathbf{m}_2$ (end) | $n_z$ switches | First switch (ps) | Settled (ps) | $f(m_{1x})$ (GHz) | Duration (ns) | Runtime (s) |
|---|---|---|---|---|---|---|---|---|---|
| r0001 | FM → AFM, A0 = -0.248e-12 | FM → AFM | -1 | 0 | – | – | – | 0.0499 | 1.32 |
| r0002 | AFM → FM, A0 = +0.248e-12 | AFM → FM | 1 | 8 | 3.7 | – | – | 0.0499 | 1.34 |

**Parameters shared by all runs (differing from the defaults)**

| Parameter | Value |
|---|---|
| m1 | 0.25, 0, 0.968 |
| t | 5e-11 |

**Parameters that differ between runs**

| Parameter | r0001 | r0002 |
|---|---|---|
| A0 | -2.48e-13 | 2.48e-13 |
| m2 | 0.35, 0, 0.937 | -0.35, 0, -0.937 |

**Discussion**

## Figures

IEEE, Nature and science styles, as PDF, SVG and PNG.

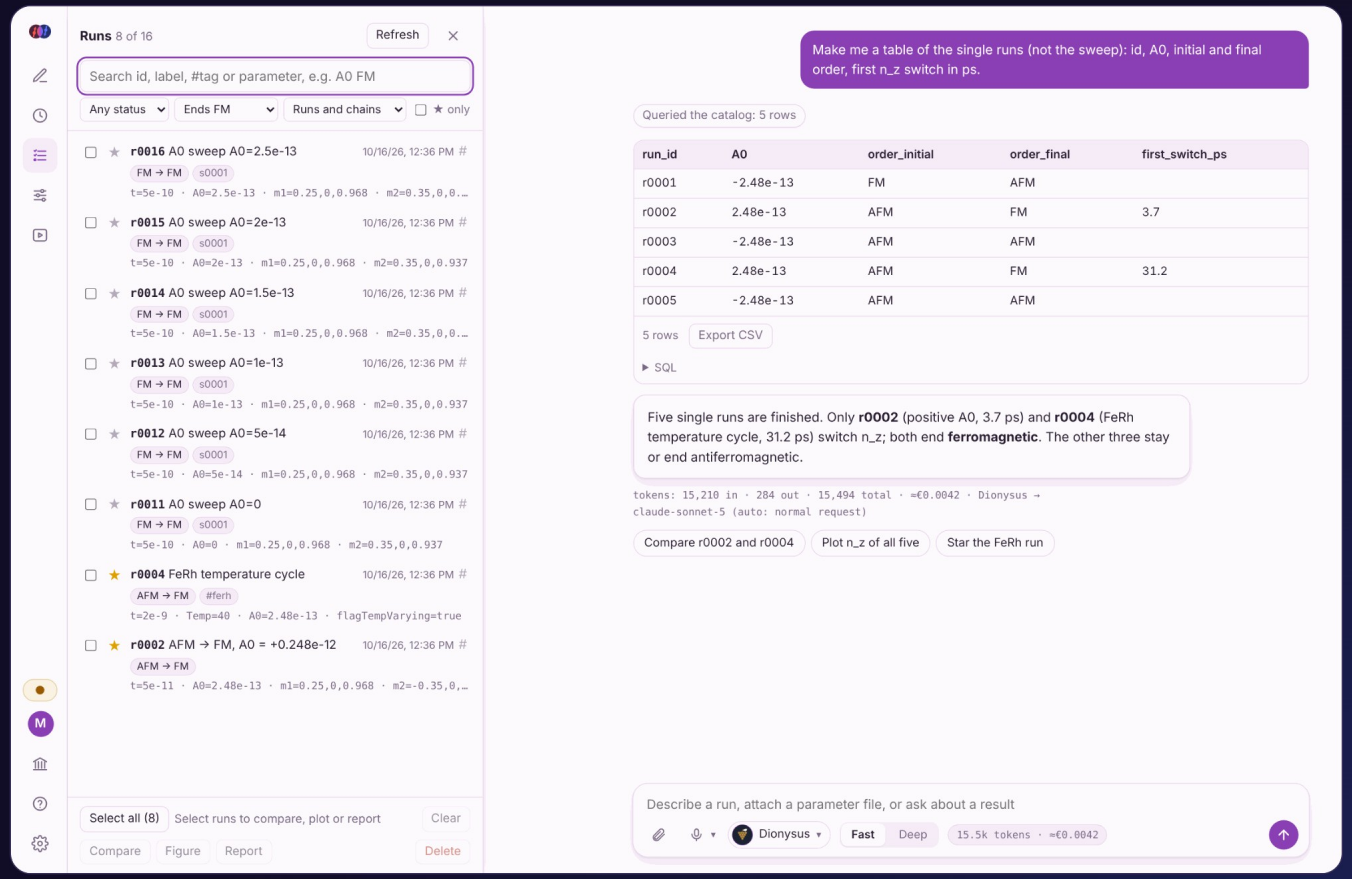
## Reports

Method, results, parameters and figures, with LaTeX math and the SpinMate header.

## Export

Print to PDF, or download the HTML or a Markdown and figures zip.

# Every run in one catalog




## One row per run

Postgres in Docker, SQLite in the Windows app: parameters, summary and the links between runs.

## Questions across runs

The model writes one read-only SELECT; the result comes back as a table with Export CSV.

## Runs panel

Search by id, label, #tag or parameter; filter, star, compare, plot or report.

# SpinMate in numbers

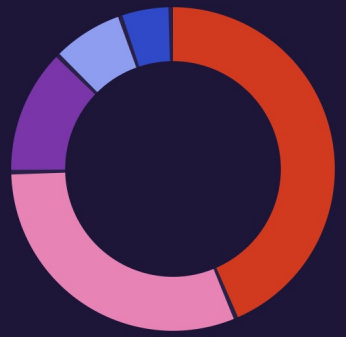
**1.3 s**
to simulate 50 ps (500 steps), run r0001

**51 s**
for an 11-run A0 sweep, 0.5 ns each

**47**
automated tests, all passing

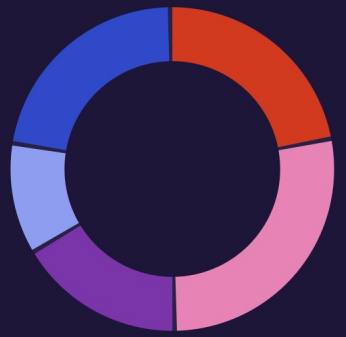
**16**
eval cases through the real agent and solver

### Codebase by part · 8,881 lines



- **44 %** agent and server
- **31 %** web UI
- **13 %** tests and evals
- **7 %** original solver, unchanged
- **5 %** desktop app and guide

### The 18 tools by job



- **4** inputs and validation
- **5** runs
- **3** sweeps
- **2** continue and combine
- **4** compare, figure, report, SQL

Measured on the runs in this deck; lines of .py, .html and .js in the repository

# Runs on your machine

### Local by default

Listens on 127.0.0.1 only. Docker on Linux, macOS and WSL, or a Windows app.

### Your keys

In .env or the browser tab, never committed. Calls bill your own account; TOKEN_BUDGET sets a cap.

### Your data

Runs, figures, reports and accounts stay in /data. Email goes out only if you set SMTP.

### Guard rails

Read-only SQL with a timeout; runs start only through the solver’s parser; big requests ask first.

# Get started

## Docker

```
cp .env.example .env
docker compose up --build
```

Add an Anthropic or OpenAI key to .env, then open localhost:8421.

## Windows app

Download SpinMate-windows.zip from Releases, unzip and run SpinMate.exe. No Docker or Python needed; the catalog is SQLite.

## First start

Athena asks your name and, optionally, your email for the PDF guide. No key yet? Run setup runs a preset directly.

spinmate **github.com/Michaellianeris/sublattice-agent**

Free and open source under the MIT license · the user guide is web/spinmate-guide.pdf